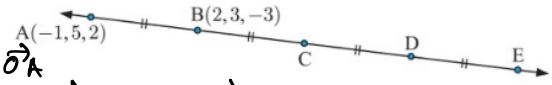
20 Find the coordinates of C, D, and E.



12I

$\vec{OB} - \vec{OA}$

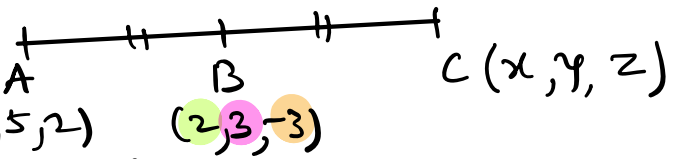
$$\vec{AB} = \begin{pmatrix} 2-(-1) \\ 3-5 \\ -3-2 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \\ -5 \end{pmatrix}$$

$$\vec{BC} = \begin{pmatrix} 2+3 \\ 3-2 \\ -3-5 \end{pmatrix} = \begin{pmatrix} 5 \\ 1 \\ -8 \end{pmatrix} \quad \therefore C(5, 1, -8)$$

$$\vec{CD} = \begin{pmatrix} 5+3 \\ 1-2 \\ -8-5 \end{pmatrix} = \begin{pmatrix} 8 \\ -1 \\ -13 \end{pmatrix} \quad \therefore D(8, -1, -13)$$

$$\vec{DE} = \begin{pmatrix} 8+3 \\ -1-2 \\ -13-5 \end{pmatrix} = \begin{pmatrix} 11 \\ -3 \\ -18 \end{pmatrix} \quad \therefore E(11, -3, -18)$$

Method (2): $A(-1, 5, 2)$, $B(2, 3, -3)$



Using midpoint formula.

$$2 = \frac{-1+x}{2} \Rightarrow 4 = -1+x \Rightarrow x = 5$$

$$3 = \frac{5+y}{2} \Rightarrow 6 = 5+y \Rightarrow y = 1$$

$$-3 = \frac{2+z}{2} \Rightarrow -6 = 2+z \Rightarrow z = -8$$

$\therefore C(5, 1, -8)$

21 A sphere has centre C(−1, 2, 4) and diameter [AB] where A is (−2, 1, 3).
12I Find the coordinates of B and the radius of the sphere.

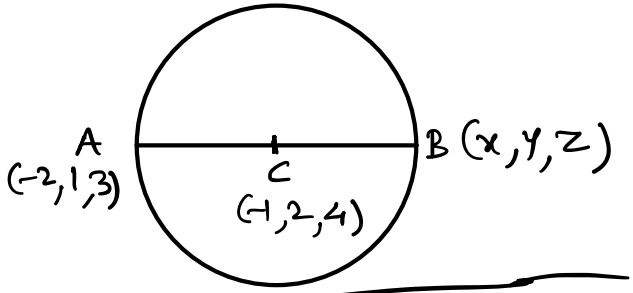
$\frac{-2+x}{2} = -1$

$-2+x = -2 \Rightarrow x = 0$

$\frac{1+y}{2} = 2 \Rightarrow y = 3$

$\frac{3+z}{2} = 4 \Rightarrow z = 5$

$\therefore B(0,3,5)$



$AC = \sqrt{(-2+1)^2+(1-2)^2+(3-4)^2}$

$= \sqrt{1+1+1}$

$= \sqrt{3}$ units.

22 Consider A$(-1, 2, 5)$, B$(2, 0, 3)$, and C$(-3, 1, 0)$.

12I

a Find the displacement vector $\overrightarrow{AB}$ and its length AB.

b Find the displacement vector $\overrightarrow{AC}$ and its length AC.

c Find the displacement vector $\overrightarrow{CB}$ and its length CB.

d Hence classify triangle ABC.

(a) $\overrightarrow{AB} = \begin{pmatrix} 2-(-1) \\ 0-2 \\ 3-5 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \\ -2 \end{pmatrix} \Rightarrow |\overrightarrow{AB}| = \sqrt{9+4+4} = \sqrt{17}$

(b) $\overrightarrow{AC} = \begin{pmatrix} -3+1 \\ 1-2 \\ 0-5 \end{pmatrix} = \begin{pmatrix} -2 \\ -1 \\ -5 \end{pmatrix} \Rightarrow |\overrightarrow{AC}| = \sqrt{4+1+25} = \sqrt{30}$

(c) $\overrightarrow{CB} = \begin{pmatrix} 2+3 \\ 0-1 \\ 3-0 \end{pmatrix} = \begin{pmatrix} 5 \\ -1 \\ 3 \end{pmatrix} \Rightarrow |\overrightarrow{CB}| = \sqrt{25+1+9} = \sqrt{35}$

$AB^2 = 17, \quad AC^2 = 30, \quad CB^2 = 35$

$\therefore \triangle ABC$ is scalene.

23 Show that $P(0, 4, 4)$, $Q(2, 6, 5)$, and $R(1, 4, 3)$ are vertices of an isosceles triangle.

12T

H.W.

24 Use side lengths to classify triangle ABC given the coordinates:

12I

a $A(0, 0, 3)$, $B(2, 8, 1)$, and $C(-9, 6, 18)$

b $A(1, 0, -3)$, $B(2, 2, 0)$, and $C(4, 6, 6)$.

25 The vertices of triangle ABC are $A(5, 6, -2)$, $B(6, 12, 9)$, and $C(2, 4, 2)$.

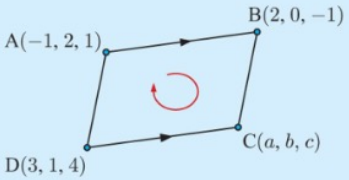
12I **a** Show that the triangle is right angled. **b** Hence find the area of the triangle.

Example 24 Self Tutor

ABCD is a parallelogram. A is $(-1, 2, 1)$, B is $(2, 0, -1)$, and D is $(3, 1, 4)$. Find the coordinates of C.

Let C be (a, b, c).

[AB] is parallel to [DC], and they have the same length, so $\overrightarrow{DC} = \overrightarrow{AB}$



$$\therefore \begin{pmatrix} a-3 \\ b-1 \\ c-4 \end{pmatrix} = \begin{pmatrix} 2-(-1) \\ 0-2 \\ -1-1 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \\ -2 \end{pmatrix}$$

$\therefore\ a-3=3,\quad b-1=-2,\quad$ and $\quad c-4=-2$

$\therefore\ a=6,\qquad b=-1,\quad$ and $\qquad c=2$

So, C is $(6, -1, 2)$.

Check: The midpoint of [DB] is $\left(\frac{3+2}{2}, \frac{1+0}{2}, \frac{4+-1}{2}\right)$ which is $\left(\frac{5}{2}, \frac{1}{2}, \frac{3}{2}\right)$.

The midpoint of [AC] is $\left(\frac{-1+6}{2}, \frac{2+-1}{2}, \frac{1+2}{2}\right)$ which is $\left(\frac{5}{2}, \frac{1}{2}, \frac{3}{2}\right)$.

The midpoints are the same, so the diagonals of the parallelogram bisect. ✓

26 A$(-1, 3, 4)$, B$(2, 5, -1)$, C$(-1, 2, -2)$, and D(r, s, t) are four points in space.

12I Find r, s, and t if:

a $\overrightarrow{AC} = \overrightarrow{BD}$ **b** $\overrightarrow{AB} = \overrightarrow{DC}$

27 A quadrilateral has vertices A(1, 2, 3), B(3, −3, 2), C(7, −4, 5), and D(5, 1, 6).

12I

a Find $\overrightarrow{AB}$ and $\overrightarrow{DC}$.

b What can be deduced about the quadrilateral ABCD?

28 PQRS is a parallelogram. P is $(-1, 2, 3)$, Q is $(1, -2, 5)$, and R is $(0, 4, -1)$.

12I

a Use vectors to find the coordinates of S.

b Use the midpoints of the diagonals to check your answer.

29 Use vectors to determine whether ABCD is a parallelogram:

12I

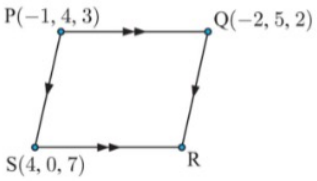
a A$(5, 0, 3)$, B$(-1, 2, 4)$, C$(4, -3, 6)$, and D$(10, -5, 5)$

b A$(2, -3, 2)$, B$(1, 4, -1)$, C$(-2, 6, -2)$, and D$(-1, -1, 2)$.

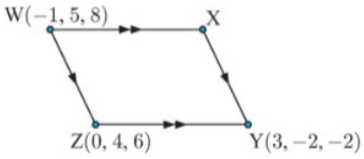
30 Use vector methods to find the remaining vertex of:

12I

a



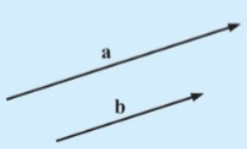
b



J PARALLELISM

Two non-zero vectors are **parallel** if and only if one is a scalar multiple of the other.

Given any non-zero vector $\mathbf{v}$ and non-zero scalar k, the vector $k\mathbf{v}$ is parallel to $\mathbf{v}$.



- If $\mathbf{a}$ is parallel to $\mathbf{b}$, then there exists a scalar k such that $\mathbf{a} = k\mathbf{b}$.
- If $\mathbf{a} = k\mathbf{b}$ for some scalar k, then
  - $\mathbf{a}$ is parallel to $\mathbf{b}$, and
  - $|\mathbf{a}| = |k||\mathbf{b}|$.

$|k|$ is the absolute value of k, whereas $|\mathbf{a}|$ is the length of vector $\mathbf{a}$.

$\vec{a} = k\vec{b}$, k is scalar.

then $\vec{a} \parallel \vec{b}$

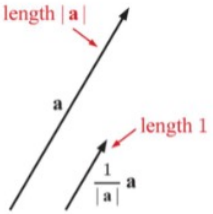
$|\vec{a}| = |k||\vec{b}|$ ✓ ✓

UNIT VECTORS

Given a non-zero vector $\mathbf{a}$, its magnitude $|\mathbf{a}|$ is a scalar quantity.

If we multiply $\mathbf{a}$ by the scalar $\frac{1}{|\mathbf{a}|}$, we obtain the parallel vector $\frac{1}{|\mathbf{a}|}\mathbf{a}$.

The length of this vector is $\left|\frac{1}{|\mathbf{a}|}\right||\mathbf{a}| = \frac{|\mathbf{a}|}{|\mathbf{a}|} = 1$, so $\frac{1}{|\mathbf{a}|}\mathbf{a}$ is a unit vector in the direction of $\mathbf{a}$. We write this unit vector as $\hat{\mathbf{a}}$.



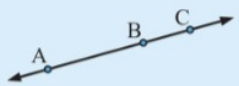
- A unit vector in the direction of $\mathbf{a}$ is $\hat{\mathbf{a}} = \frac{1}{|\mathbf{a}|}\mathbf{a}$.
- A vector $\mathbf{b}$ of length k in the same direction as $\mathbf{a}$ is $\mathbf{b} = k\hat{\mathbf{a}} = \frac{k}{|\mathbf{a}|}\mathbf{a}$.
- A vector $\mathbf{b}$ of length k which is *parallel to* $\mathbf{a}$ could be $\mathbf{b} = \pm k\hat{\mathbf{a}} = \pm\frac{k}{|\mathbf{a}|}\mathbf{a}$.

i, j, k are base unit vectors.

COLLINEAR POINTS

Three or more points are said to be **collinear** if they lie on the same straight line.

A, B, and C are **collinear** if $\overrightarrow{AB} = k\overrightarrow{BC}$ for some scalar k.



Example 27 **Self Tutor**

Prove that $A(-1, 2, 3)$, $B(4, 0, -1)$, and $C(14, -4, -9)$ are collinear.

$$\overrightarrow{AB} = \begin{pmatrix} 5 \\ -2 \\ -4 \end{pmatrix} \text{ and } \overrightarrow{BC} = \begin{pmatrix} 10 \\ -4 \\ -8 \end{pmatrix} = 2\begin{pmatrix} 5 \\ -2 \\ -4 \end{pmatrix}$$

$\therefore \overrightarrow{BC} = 2\overrightarrow{AB}$

$\therefore$ [BC] is parallel to [AB].

Since B is common to both, A, B, and C are collinear.

EXERCISE 12J

1 Determine whether each pair of vectors is parallel:

a $\begin{pmatrix}2\\5\end{pmatrix}$ and $\begin{pmatrix}6\\15\end{pmatrix}$ b $\begin{pmatrix}1\\-4\end{pmatrix}$ and $\begin{pmatrix}2\\-10\end{pmatrix}$ c $\begin{pmatrix}6\\-12\end{pmatrix}$ and $\begin{pmatrix}-4\\8\end{pmatrix}$

d $\begin{pmatrix}1\\4\\-3\end{pmatrix}$ and $\begin{pmatrix}-2\\-8\\6\end{pmatrix}$ e $\begin{pmatrix}8\\0\\-20\end{pmatrix}$ and $\begin{pmatrix}-6\\0\\15\end{pmatrix}$ f $\begin{pmatrix}6\\10\\-2\end{pmatrix}$ and $\begin{pmatrix}24\\40\\8\end{pmatrix}$

(a) $\vec{a} = \begin{pmatrix}2\\5\end{pmatrix} \rightarrow \vec{b} \begin{pmatrix}6\\15\end{pmatrix} = 3\begin{pmatrix}2\\5\end{pmatrix}$ parallel

$\vec{b} = 3\vec{a}$

(c) $3\begin{pmatrix}2\\-4\end{pmatrix}$ $\begin{pmatrix}-4\\8\end{pmatrix} = -2\begin{pmatrix}2\\-4\end{pmatrix}$ parallel

(e) parallel.

2 Find a such that:

12J **a** $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} -6 \\ a \end{pmatrix}$ are parallel **b** $\begin{pmatrix} a \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ -1 \end{pmatrix}$ are parallel.

$-3 \times \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} -6 \\ 3 \end{pmatrix} = \begin{pmatrix} -6 \\ a \end{pmatrix} \Rightarrow a = 3$

(b) $-2 \times \begin{pmatrix} 3 \\ -1 \end{pmatrix} = \begin{pmatrix} -6 \\ 2 \end{pmatrix} = \begin{pmatrix} a \\ 2 \end{pmatrix} \Rightarrow a = -6$

3 Find r and s given that $\mathbf{a}$ and $\mathbf{b}$ are parallel:

12J

a $\mathbf{a} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -6 \\ r \\ s \end{pmatrix}$

b $\mathbf{a} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} r \\ 2 \\ s \end{pmatrix}$

4 What can be deduced from the following?

12J

a $\overrightarrow{AB} = 3\overrightarrow{CD}$ **b** $\overrightarrow{RS} = -\frac{1}{2}\overrightarrow{KL}$ **c** $\overrightarrow{AB} = 2\overrightarrow{BC}$

5 The position vectors of P, Q, R, and S are $\begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix}$, $\begin{pmatrix} 1 \\ 4 \\ -3 \end{pmatrix}$, $\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$, and $\begin{pmatrix} -1 \\ -2 \\ 3 \end{pmatrix}$ respectively.

12J

Show that [PR] and [QS] are parallel, and state the ratio PR : QS.

6 If $\mathbf{a} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$, write down the vector:

12J

a in the same direction as $\mathbf{a}$ and with twice its length

b in the opposite direction to $\mathbf{a}$ and with half its length.

7 Find the 2-dimensional unit vector in the direction:

12点

a $\mathbf{i} + 2\mathbf{j}$ **b** $2\mathbf{i} - 3\mathbf{j}$ **c** $-2\mathbf{i} + 2\mathbf{j}$

8 Find the 3-dimensional unit vector in the direction:

12J

a $2\mathbf{i} + 3\mathbf{k}$ **b** $-\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ **c** $2\mathbf{i} - 2\mathbf{j} + \mathbf{k}$

9 Find a vector $\mathbf{v}$ which has:

12J

a the same direction as $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ and with length 3 units

b the opposite direction to $\begin{pmatrix} -1 \\ -4 \end{pmatrix}$ and with length 2 units

c the same direction as $\begin{pmatrix} -1 \\ 4 \\ 1 \end{pmatrix}$ and with length 6 units

d the opposite direction to $-\mathbf{i} - 2\mathbf{j} - 2\mathbf{k}$ and with length 5 units.

10 **a** Find the vectors of length 1 unit which are parallel to $\mathbf{a} = \begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix}$.

12J

b Find vectors of length 2 units which are parallel to $\mathbf{b} = \begin{pmatrix} -2 \\ -1 \\ 2 \end{pmatrix}$.

(a) $\vec{a} = \begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix} \Rightarrow |\vec{a}| = \sqrt{4+1+4} = 3$

$\therefore$ unit vector parallel to $\vec{a} = \dfrac{\vec{a}}{|\vec{a}|}$

$= \dfrac{\begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix}}{3}$

$= \begin{pmatrix} 2/3 \\ -1/3 \\ -2/3 \end{pmatrix}$

$\mathbf{b} = \begin{pmatrix} -2 \\ -1 \\ 2 \end{pmatrix} \Rightarrow |\vec{b}| = \sqrt{4+1+4} = 3$

$\therefore$ unit vector parallel to $\vec{b} = \dfrac{\vec{b}}{|\vec{b}|} = \dfrac{1}{3}\begin{pmatrix} -2 \\ -1 \\ 2 \end{pmatrix}$

11 A is $(3, 2)$ and point B is 4 units from A in the direction $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

12J

a Find $\overrightarrow{AB}$.

b Find $\overrightarrow{OB}$ using $\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB}$.

c Hence deduce the coordinates of B.

12 Find the coordinates of point X which is:

12J

a 5 units from $(1, -5)$ in the direction $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$

b 6 units from $(1, 3, -2)$ in the direction $\begin{pmatrix} -3 \\ 0 \\ 4 \end{pmatrix}$

c 4 units from $(2, -1, 4)$ in the direction $\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$.

(a) $\left|\begin{pmatrix} 3 \\ 4 \end{pmatrix}\right| = \sqrt{9+16} = 5$ units

$$\begin{pmatrix} 1 \\ -5 \end{pmatrix} + \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \end{pmatrix}$$

∴ a point 5 units from $(1, -5)$ in direction of $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ is $(4, -1)$

✓

b 6 units from $(1, 3, -2)$ in the direction $\begin{pmatrix} -3 \\ 0 \\ 4 \end{pmatrix}$

$\left|\begin{pmatrix} -3 \\ 0 \\ 4 \end{pmatrix}\right| = \sqrt{9+16} = 5$

unit vector in direction of $\begin{pmatrix} -3 \\ 0 \\ 4 \end{pmatrix} = \frac{1}{5}\begin{pmatrix} -3 \\ 0 \\ 4 \end{pmatrix}$

∴ a vector of 6 units is $\frac{6}{5}\begin{pmatrix} -3 \\ 0 \\ 4 \end{pmatrix}$

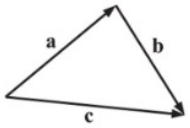
$$\therefore \quad \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix} + \frac{6}{5}\begin{pmatrix} -3 \\ 0 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 - 18/5 \\ 3 \\ -2 + \frac{24}{5} \end{pmatrix} = \begin{pmatrix} -13/5 \\ 3 \\ 14/5 \end{pmatrix}$$

$\therefore$ The point is $\left(\frac{-13}{5}, 3, \frac{14}{5}\right)$

13 The **triangle inequality** states that:

125 "In any triangle, the sum of any two sides must always be greater than the third side."

Prove that $|\mathbf{a}+\mathbf{b}| \leqslant |\mathbf{a}| + |\mathbf{b}|$ using a geometrical argument.



Hint: Consider:

- **a** not parallel to **b** and use the triangle inequality
- **a** and **b** parallel
- any other cases.

14 Show that the following sets of points are collinear:

12J

a A(−1, 1), B(4, 6), C(1, 3)

b P(3, 3), Q(6, −3), R(1, 7)

c A(−2, 1, 4), B(4, 3, 0), C(19, 8, −10)

d P(2, 1, 1), Q(5, −5, −2), R(−1, 7, 4)

(a) $\overrightarrow{AB} = \begin{pmatrix} 5 \\ 5 \end{pmatrix} = 5\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ $\quad \frac{1}{5}\overrightarrow{AB} = -\frac{1}{3}\overrightarrow{BC}$

$\overrightarrow{BC} = \begin{pmatrix} -3 \\ -3 \end{pmatrix} = -3\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$\therefore$ A, B, C are collinear.

15 Find a and b if:

12J **a** A$(2, -3, 4)$, B$(11, -9, 7)$, and C$(-13, a, b)$ are collinear

b K$(1, -1, 0)$, L$(4, -3, 7)$, and M$(a, 2, b)$ are collinear.

(a) $\overrightarrow{AB} = \begin{pmatrix} 9 \\ -6 \\ 3 \end{pmatrix}$ $\overrightarrow{BC} = \begin{pmatrix} -24 \\ a+9 \\ b-7 \end{pmatrix}$ or $9 = k(-24)$

$k = -\frac{9}{24} = \boxed{-\frac{3}{8}}$

$\therefore +6 = +\frac{3}{8}(a+9) \Rightarrow 16 = a+9 \Rightarrow \boxed{a = 7}$

$3 = -\frac{3}{8}(b-7) \Rightarrow -8 = b-7 \Rightarrow \boxed{b = -1}$

(b) $\overrightarrow{KL} = \begin{pmatrix} 3 \\ -2 \\ 7 \end{pmatrix}$ $\overrightarrow{LM} = \begin{pmatrix} a-4 \\ 5 \\ b-7 \end{pmatrix}$

Let $-2 = k \times 5 \Rightarrow k = -\frac{2}{5}$

$\therefore 3 = -\frac{2}{5}(a-4) \Rightarrow -\frac{15}{2} = a-4$

$a = -\frac{15}{2} + 4 = \boxed{-\frac{7}{2}}$

$\boxed{b = -\frac{21}{2}}$

16 Let $P(x, y, z)$ be a point on a sphere with radius r units and centre $C(X, Y, Z)$.

12J **a** Write down the vector $\overrightarrow{CP}$.

b Use the distance formula for CP to write a Cartesian equation for the sphere.

(a) $\overrightarrow{CP} = \begin{pmatrix} x-X \\ y-Y \\ z-Z \end{pmatrix}$

(b) $|\overrightarrow{CP}| = \sqrt{(x-X)^2+(y-Y)^2+(z-Z)^2}$

$\therefore \sqrt{(x-X)^2+(y-Y)^2+(z-Z)^2} = r$

$(x-X)^2+(y-Y)^2+(z-Z)^2 = r^2$

K THE SCALAR PRODUCT OF TWO VECTORS

For ordinary numbers a and b we can write the product of a and b as ab or $a \times b$. There is only one interpretation for this product, so we can use power notation $a^2 = a \times a$, $a^3 = a \times a \times a$, and so on as shorthand.

However, there are *two* different types of product involving two vectors. These are:

- The **scalar product** of 2 vectors, which results in a **scalar** answer and has the notation **a** • **b** (read "**a** dot **b**").
- The **vector product** of 2 vectors, which results in a **vector** answer and has the notation **a** × **b** (read "**a** cross **b**").

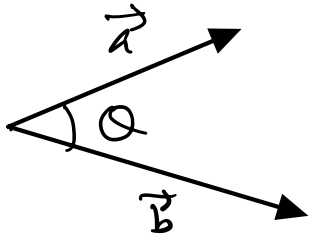
$\vec{a} = 2i + 3j + 4k \rightarrow |\vec{a}| = \sqrt{4+9+16} = \sqrt{29}$

$\vec{b} = i - 2j + 3k \rightarrow |\vec{b}| = \sqrt{1+4+9} = \sqrt{14}$

$\therefore \quad \vec{a} \cdot \vec{b} = (2)(1) + (3)(-2) + (4)(3) = 2 - 6 + 12 = \boxed{8}$

Then

$$\boxed{\vec{a} \cdot \vec{b} = |\vec{a}|\,|\vec{b}| \cos\theta}$$



$\therefore \quad \cos\theta = \dfrac{|\vec{a} \cdot \vec{b}|}{|\vec{a}|\,|\vec{b}|}$ ← for acute angle.

$\therefore \quad \cos\theta = \dfrac{8}{\sqrt{29}\sqrt{14}} \Rightarrow \theta = \cos^{-1}\left(\dfrac{8}{\sqrt{29 \times 14}}\right) \approx \boxed{66.6^\circ}$

Consequently, for vector **a**, we do not use the terms $\mathbf{a}^2$ or $(\mathbf{a})^2$, as it is not clear which of the vector products it refers to.

For 2-dimensional vectors in a plane, we only need to talk about scalar product. The vector cross product will be important for 3-dimensional vectors in space.

SCALAR PRODUCT

The **scalar product** of two vectors is also known as the **dot product** or **inner product**.

► $\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} \bullet \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = a_1b_1 + a_2b_2$

► $\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \bullet \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = a_1b_1 + a_2b_2 + a_3b_3$

Be careful not to confuse the **scalar product**, which is the product of two vectors to give a scalar answer, with **scalar multiplication**, which is the product of a scalar and a vector to give a parallel vector. They are quite different.

ALGEBRAIC PROPERTIES OF THE SCALAR PRODUCT

1 $\mathbf{a} \bullet \mathbf{b} = \mathbf{b} \bullet \mathbf{a}$

2 $\mathbf{a} \bullet \mathbf{a} = |\mathbf{a}|^2$

3 $\mathbf{a} \bullet (\mathbf{b} + \mathbf{c}) = \mathbf{a} \bullet \mathbf{b} + \mathbf{a} \bullet \mathbf{c}$

4 $(\mathbf{a} + \mathbf{b}) \bullet (\mathbf{c} + \mathbf{d}) = \mathbf{a} \bullet \mathbf{c} + \mathbf{a} \bullet \mathbf{d} + \mathbf{b} \bullet \mathbf{c} + \mathbf{b} \bullet \mathbf{d}$

EXERCISE 12K

1 Find:

a $\begin{pmatrix}2\\5\end{pmatrix}\bullet\begin{pmatrix}3\\1\end{pmatrix}$ b $\begin{pmatrix}5\\-1\end{pmatrix}\bullet\begin{pmatrix}2\\4\end{pmatrix}$ c $\begin{pmatrix}3\\0\\-2\end{pmatrix}\bullet\begin{pmatrix}4\\-2\\5\end{pmatrix}$ d $\begin{pmatrix}-4\\-5\\3\end{pmatrix}\bullet\begin{pmatrix}9\\-6\\-1\end{pmatrix}$

(a) $\begin{pmatrix}2\\5\end{pmatrix}\bullet\begin{pmatrix}3\\1\end{pmatrix}=(2)(3)+(5)(1)=6+5=11$

(c) $\begin{pmatrix}3\\0\\-2\end{pmatrix}\bullet\begin{pmatrix}4\\-2\\5\end{pmatrix}=(3)(4)+(0)+(-2)(5)=12-10=2$

2 For $\mathbf{p} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$, $\mathbf{q} = \begin{pmatrix} -1 \\ 5 \end{pmatrix}$, and $\mathbf{r} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$, find:

12K

a $\mathbf{q} \bullet \mathbf{p}$ **b** $\mathbf{q} \bullet \mathbf{r}$ **c** $\mathbf{q} \bullet (\mathbf{p} + \mathbf{r})$ **d** $3\mathbf{r} \bullet \mathbf{q}$

e $2\mathbf{p} \bullet 2\mathbf{p}$ **f** $\mathbf{i} \bullet \mathbf{p}$ **g** $\mathbf{q} \bullet \mathbf{j}$ **h** $\mathbf{i} \bullet \mathbf{i}$

(e) $\vec{a} \cdot \vec{a} = |\vec{a}|^2$

$2\vec{p} \cdot 2\vec{p} = |2\vec{p}|^2 = |2|\,|\vec{p}|^2 = 4\left(\sqrt{3^2+2^2}\right)^2 = 4 \times 13 = 52$

3 For $\mathbf{a} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$, and $\mathbf{c} = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$, find:

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a $\mathbf{a} \bullet \mathbf{b}$ **b** $\mathbf{b} \bullet \mathbf{a}$ **c** $|\mathbf{a}|^2$ ✓

d $\mathbf{a} \bullet \mathbf{a}$ **e** $\mathbf{a} \bullet (\mathbf{b} + \mathbf{c})$ **f** $\mathbf{a} \bullet \mathbf{b} + \mathbf{a} \bullet \mathbf{c}$

(c) $|\vec{a}|^2 = \vec{a} \cdot \vec{a} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} = 4 + 1 + 9 = 14$

4 Use the general vectors $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ to prove that:

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a $\mathbf{a} \bullet \mathbf{b} = \mathbf{b} \bullet \mathbf{a}$ **b** $\mathbf{a} \bullet \mathbf{a} = |\mathbf{a}|^2$

5 Find:

12K

a $(\mathbf{i}+\mathbf{j}-\mathbf{k})\bullet(2\mathbf{j}+\mathbf{k})$ **b** $\mathbf{i}\bullet\mathbf{i}$ **c** $(3\mathbf{i}-2\mathbf{k})\bullet(\mathbf{i}+\mathbf{j})$

d $(2\mathbf{i}+\mathbf{k})\bullet(-\mathbf{i}+3\mathbf{j}+2\mathbf{k})$ **e** $\mathbf{i}\bullet\mathbf{j}$ **f** $(\mathbf{i}-4\mathbf{j}+2\mathbf{k})\bullet(3\mathbf{i}-3\mathbf{j}-5\mathbf{k})$

(a) $\begin{pmatrix}1\\1\\-1\end{pmatrix}\cdot\begin{pmatrix}0\\2\\1\end{pmatrix} = 0+2-1 = 1$

6 Let $\mathbf{a} = \begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$, $\mathbf{c} = \begin{pmatrix} 3 \\ -1 \\ 5 \end{pmatrix}$, and $\mathbf{d} = \begin{pmatrix} 0 \\ -2 \\ 3 \end{pmatrix}$.

12K

a Find:

i $\mathbf{a} \bullet \mathbf{c}$ **ii** $\mathbf{a} \bullet \mathbf{d}$ **iii** $\mathbf{b} \bullet \mathbf{c}$ **iv** $\mathbf{b} \bullet \mathbf{d}$

b Hence show that $(\mathbf{a} + \mathbf{b}) \bullet (\mathbf{c} + \mathbf{d}) = \mathbf{a} \bullet \mathbf{c} + \mathbf{a} \bullet \mathbf{d} + \mathbf{b} \bullet \mathbf{c} + \mathbf{b} \bullet \mathbf{d}$.

7 Use $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$, and $\mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$ to prove that $\mathbf{a} \bullet (\mathbf{b} + \mathbf{c}) = \mathbf{a} \bullet \mathbf{b} + \mathbf{a} \bullet \mathbf{c}$.

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Hence prove that $(\mathbf{a} + \mathbf{b}) \bullet (\mathbf{c} + \mathbf{d}) = \mathbf{a} \bullet \mathbf{c} + \mathbf{a} \bullet \mathbf{d} + \mathbf{b} \bullet \mathbf{c} + \mathbf{b} \bullet \mathbf{d}$.

8 Use the properties of the scalar product to show that:

12k

a $|\mathbf{a}+\mathbf{b}|^2 = |\mathbf{a}|^2 + 2\mathbf{a} \bullet \mathbf{b} + |\mathbf{b}|^2$

b $|\mathbf{a}+\mathbf{b}|^2 + |\mathbf{a}-\mathbf{b}|^2 = 2|\mathbf{a}|^2 + 2|\mathbf{b}|^2$

c $|\mathbf{a}+\mathbf{b}|^2 - |\mathbf{a}-\mathbf{b}|^2 = 4\mathbf{a} \bullet \mathbf{b}$

12K

9 Explain why $\mathbf{a} \bullet \mathbf{b} \bullet \mathbf{c}$ is meaningless.

10 If $|\mathbf{a}| = 5$ and $|\mathbf{b}| = 3$, find $(\mathbf{a} + \mathbf{b}) \bullet (\mathbf{a} - \mathbf{b})$.

(9) $\vec{a} \cdot \vec{b}$ is a scalar

Now scalar cannot have dot product with c

∴ it is meaningless

10 If $|\mathbf{a}| = 5$ and $|\mathbf{b}| = 3$, find $(\mathbf{a} + \mathbf{b}) \bullet (\mathbf{a} - \mathbf{b})$.

$(\vec{a}+\vec{b}) \cdot (\vec{a}-\vec{b})$

$= \vec{a} \cdot \vec{a} - \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{a} - \vec{b} \cdot \vec{b}$

$= |\vec{a}|^2 + 0 - |\vec{b}|^2$

$= 5^2 - 3^2 = 25 - 9 = \boxed{16}$

11 Show that if vectors **a** and **b** are parallel, then $|\mathbf{a} \bullet \mathbf{b}| = |\mathbf{a}||\mathbf{b}|$.

12K

If $\vec{a} \parallel \vec{b} \Rightarrow \vec{b} = k \cdot \vec{a}$

$\therefore |\vec{a} \cdot \vec{b}| = |\vec{a} \cdot k\vec{a}| = |k|\,|\vec{a} \cdot \vec{a}| = |k||a|^2$

$= |k|\,|a|\,|\vec{a}|$

$= |\vec{a}| \times |k \cdot \vec{a}|$

$\therefore |\vec{a} \cdot \vec{b}| = |\vec{a}| \times |\vec{b}|$

L THE ANGLE BETWEEN TWO VECTORS

DISCUSSION

1



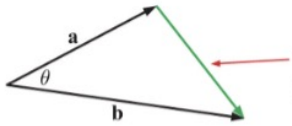
Elaine has drawn a vector **v** on a *plane* which is a sheet of paper. It is therefore a 2-dimensional vector.

a How many vectors can she draw which are *perpendicular* to **v**?

b Are all of these vectors parallel?

THE ANGLE BETWEEN TWO VECTORS

Suppose θ is the angle between the vectors $\mathbf{a}$ and $\mathbf{b}$.



This vector is $-\mathbf{a} + \mathbf{b}$ or $\mathbf{b} - \mathbf{a}$.
Its length is $|\mathbf{b} - \mathbf{a}|$.

Using the cosine rule, $|\mathbf{b} - \mathbf{a}|^2 = |\mathbf{a}|^2 + |\mathbf{b}|^2 - 2|\mathbf{a}||\mathbf{b}|\cos\theta$

Letting $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$,

$$\mathbf{b} - \mathbf{a} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} - \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} b_1 - a_1 \\ b_2 - a_2 \\ b_3 - a_3 \end{pmatrix}$$

$$\therefore \ (b_1 - a_1)^2 + (b_2 - a_2)^2 + (b_3 - a_3)^2 = (a_1^2 + a_2^2 + a_3^2) + (b_1^2 + b_2^2 + b_3^2) - 2|\mathbf{a}||\mathbf{b}|\cos\theta$$

$$\therefore \ 2a_1b_1 + 2a_2b_2 + 2a_3b_3 = 2|\mathbf{a}||\mathbf{b}|\cos\theta \qquad \{\text{simplifying}\}$$

$$\therefore \ a_1b_1 + a_2b_2 + a_3b_3 = |\mathbf{a}||\mathbf{b}|\cos\theta$$

$$\therefore \ \mathbf{a} \bullet \mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta$$

The angle θ between two vectors $\mathbf{a}$ and $\mathbf{b}$ can be found using $\cos\theta = \dfrac{\mathbf{a} \bullet \mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$.

GEOMETRIC PROPERTIES OF THE SCALAR PRODUCT

Using the formula $\mathbf{a} \bullet \mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta$, we notice that:

- If $\mathbf{a}$ is perpendicular to $\mathbf{b}$ then $\theta = 90°$.

$$\begin{aligned} \therefore \quad \mathbf{a} \bullet \mathbf{b} &= |\mathbf{a}||\mathbf{b}|\cos\theta \\ &= |\mathbf{a}||\mathbf{b}|\cos 90° \\ &= 0 \end{aligned}$$

- If $\mathbf{a}$ is parallel to $\mathbf{b}$ then $\theta = 0°$ or $180°$.

$$\begin{aligned} \therefore \quad \mathbf{a} \bullet \mathbf{b} &= |\mathbf{a}||\mathbf{b}|\cos\theta \\ &= |\mathbf{a}||\mathbf{b}|\cos 0° \quad \text{or} \quad |\mathbf{a}||\mathbf{b}|\cos 180° \\ &= \pm|\mathbf{a}||\mathbf{b}| \\ \therefore \quad |\mathbf{a} \bullet \mathbf{b}| &= |\mathbf{a}||\mathbf{b}| \end{aligned}$$

We are hence able to state the following **geometric properties of the scalar product**:

For non-zero vectors $\mathbf{a}$ and $\mathbf{b}$:

1 $\mathbf{a} \bullet \mathbf{b} = 0 \quad \Leftrightarrow \quad \mathbf{a}$ and $\mathbf{b}$ are **perpendicular**.

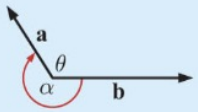
2 $|\mathbf{a} \bullet \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \quad \Leftrightarrow \quad \mathbf{a}$ and $\mathbf{b}$ are **parallel vectors**.

3 If θ is the angle between vectors $\mathbf{a}$ and $\mathbf{b}$ then $\mathbf{a} \bullet \mathbf{b} = |\mathbf{a}||\mathbf{b}|\cos\theta$.

If θ is acute, $\cos\theta > 0$ and so $\mathbf{a} \bullet \mathbf{b} > 0$

If θ is obtuse, $\cos\theta < 0$ and so $\mathbf{a} \bullet \mathbf{b} < 0$.

The angle between two vectors is always taken as the angle θ such that $0° \leqslant \theta \leqslant 180°$, rather than reflex angle α.



EXERCISE 12L

1 If $\mathbf{a} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$, find:

a $\mathbf{a} \bullet \mathbf{b}$

b the angle between $\mathbf{a}$ and $\mathbf{b}$.

2 Find the angle between $\mathbf{r}$ and $\mathbf{s}$, given:

12L

a $\mathbf{r} = \begin{pmatrix} 3 \\ 3 \end{pmatrix}$ and $\mathbf{s} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$

b $\mathbf{r} = \begin{pmatrix} -1 \\ -3 \end{pmatrix}$ and $\mathbf{s} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$

c $\mathbf{r} = \mathbf{i} - \mathbf{j}$ and $\mathbf{s} = 2\mathbf{i} - \mathbf{j}$

d $\mathbf{r} = \mathbf{i}$ and $\mathbf{s} = \mathbf{i} + \mathbf{j}$

3 Show that $\begin{pmatrix} 1 \\ 1 \\ 5 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix}$ are perpendicular.

12L

$\begin{pmatrix} 1 \\ 1 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} = 2 + 3 - 5 = 0 \Rightarrow$ They are $\perp^r$

4 Show that $\mathbf{a} = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$, and $\mathbf{c} = \begin{pmatrix} 1 \\ 5 \\ -4 \end{pmatrix}$ are mutually perpendicular.

12 L

Vectors are *mutually perpendicular* if each one is perpendicular to all the others.

$\vec{a} \cdot \vec{b} = -3 + 1 + 2 = 0$

$\vec{b} \cdot \vec{c} = -1 + 5 - 4 = 0$

$\vec{c} \cdot \vec{a} = 3 + 5 - 8 = 0$

$\therefore \vec{a} \perp \vec{b} \perp \vec{c}$

H. W.

5 Find the angle between:

12L

a $\begin{pmatrix} 4 \\ 0 \\ -2 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}$

b $\begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix}$

c $2\mathbf{j} - \mathbf{k}$ and $\mathbf{i} + 2\mathbf{k}$

d $\mathbf{j}$ and $-\mathbf{i} + \mathbf{j} + 3\mathbf{k}$

H.W.

6 Find t given that the following pairs of vectors are perpendicular:

12L

a $\mathbf{p} = \begin{pmatrix} 3 \\ t \end{pmatrix}$, $\mathbf{q} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$

b $\mathbf{r} = \begin{pmatrix} t \\ t+2 \end{pmatrix}$, $\mathbf{s} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$

c $\mathbf{a} = \begin{pmatrix} t \\ t+2 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 2-3t \\ t \end{pmatrix}$

d $\mathbf{a} = \begin{pmatrix} 3 \\ -1 \\ t \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 2t \\ -3 \\ -4 \end{pmatrix}$

e $\mathbf{p} = \begin{pmatrix} 3 \\ t \\ -2 \end{pmatrix}$, $\mathbf{q} = \begin{pmatrix} 1-t \\ -3 \\ 4 \end{pmatrix}$

f $\mathbf{a} = 2\mathbf{i} + t\mathbf{j} + (t-2)\mathbf{k}$, $\mathbf{b} = t\mathbf{i} + 3\mathbf{j} + t\mathbf{k}$

H.W.

7 Find r, s, and t such that $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 2 \\ 2 \\ r \end{pmatrix}$, and $\mathbf{c} = \begin{pmatrix} s \\ t \\ 1 \end{pmatrix}$ are mutually perpendicular.

12L

End of Lesson

8 Find a vector of length:

12L

a 10 units which is perpendicular to $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$

b $3\sqrt{2}$ units which is perpendicular to $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$

✓ **c** $\sqrt{20}$ units which is perpendicular to $\begin{pmatrix} -2 \\ -1 \end{pmatrix}$.

(a) a vector $\perp^r$ to $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$ is $\begin{pmatrix} 4 \\ 3 \end{pmatrix}$ ✓ $\rightarrow 12 - 12 = 0$

$\left|\begin{pmatrix} 4 \\ 3 \end{pmatrix}\right| = \sqrt{16+9} = 5$

$\therefore$ unit vector in the direction of $\begin{pmatrix} 4 \\ 3 \end{pmatrix} = \frac{1}{5}\begin{pmatrix} 4 \\ 3 \end{pmatrix}$

$\therefore$ vector of length 10 units $= \frac{10}{5}\begin{pmatrix} 4 \\ 3 \end{pmatrix} = \begin{pmatrix} 8 \\ 6 \end{pmatrix}$

b $3\sqrt{2}$ units which is perpendicular to $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$

Vector $\perp^r$ to $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ is $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ ✓

$\begin{pmatrix} 1 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1 - 1 = 0$

$\left|\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right| = \sqrt{2}$

$\therefore$ unit vector in direction of $\begin{pmatrix}1\\1\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}$

$\therefore$ Vector of length $3\sqrt{2}$ in $\begin{pmatrix}1\\1\end{pmatrix} = \frac{3\sqrt{2}}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}3\\3\end{pmatrix}$

9 Find *two* vectors of length 5 units which are perpendicular to $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$.

12L

Two vector ⊥r to $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ are $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$ & $\begin{pmatrix} -3 \\ 2 \end{pmatrix}$

$\begin{pmatrix} 2 \\ 3 \end{pmatrix}\begin{pmatrix} 3 \\ -2 \end{pmatrix} = 6-6=0$

$\begin{pmatrix} 2 \\ 3 \end{pmatrix}\begin{pmatrix} -3 \\ 2 \end{pmatrix} = -6+6=0$

$\left|\begin{pmatrix} 3 \\ -2 \end{pmatrix}\right| = \sqrt{9+4} = \sqrt{13}$

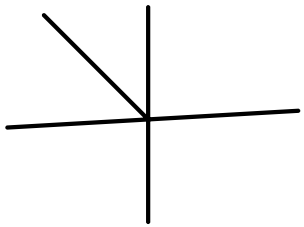
∴ Vectors of length 5 units are $\frac{5}{\sqrt{13}}\begin{pmatrix} 3 \\ -2 \end{pmatrix}$

& $\frac{5}{\sqrt{13}}\begin{pmatrix} -3 \\ 2 \end{pmatrix}$

10 $\mathbf{a}$ and $\mathbf{b}$ have lengths $\sqrt{10}$ units and 3 units respectively and the angle between them is $71°$.
12 L Find $\mathbf{a} \bullet \mathbf{b}$.

11 Find $\mathbf{p} \bullet \mathbf{q}$ for:

a $|\mathbf{p}| = 2$, $|\mathbf{q}| = 5$, and $\theta = 60°$ **b** $|\mathbf{p}| = 6$, $|\mathbf{q}| = 3$, and $\theta = \frac{2\pi}{3}$



(10) $\cos\theta = \dfrac{|\vec{a} \cdot \vec{b}|}{|\vec{a}||\vec{b}|}$

$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\theta$

$= (\sqrt{10})(3)\cos 71° = \boxed{3.09}$

(11)(a) $\vec{p} \cdot \vec{q} = |p||\vec{q}| \cos 60$

$= (2)(5)\frac{1}{2} = \boxed{5}$

(b) $\vec{p} \cdot \vec{q} = 6 \times 3 \times (-\frac{1}{2}) = \boxed{-9}$

12 Prove that two non-zero vectors **a** and **b** are perpendicular if and only if $\mathbf{a} \bullet \mathbf{b} = 0$.

12L **Hint:** You need to show that **a** and **b** being perpendicular implies $\mathbf{a} \bullet \mathbf{b} = 0$ *and* that $\mathbf{a} \bullet \mathbf{b} = 0$ implies **a** and **b** are perpendicular.

iff → if and only if.

This is two way proof.

(1) If $\vec{a} \perp \vec{b} \Rightarrow \vec{a} \cdot \vec{b} = 0$

(2) If $\vec{a} \cdot \vec{b} = 0 \Rightarrow \vec{a} \perp \vec{b}$

(1) $\vec{a} \perp \vec{b}$ $\therefore \vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos 90^\circ = 0$

$\therefore \vec{a} \cdot \vec{b} = 0$

(2) let $\vec{a} \cdot \vec{b} = 0 \Rightarrow |\vec{a}||\vec{b}| \cos\theta = 0$

$\cos\theta = 0 \Rightarrow \theta = 90^\circ$

$\therefore \vec{a} \perp \vec{b}$

iff

13 Prove that if **a** and **b** are non-zero vectors then $|\mathbf{a} \bullet \mathbf{b}| = |\mathbf{a}||\mathbf{b}| \Leftrightarrow$ **a** and **b** are parallel.

12 L

Let $\vec{a} \parallel \vec{b} \Rightarrow \theta = 0$

$\therefore \vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos 0 = |\vec{a}||\vec{b}|$

$\therefore |\vec{a} \cdot \vec{b}| = |\vec{a}||\vec{b}|$ ✓

alternatively let $|\vec{a} \cdot \vec{b}| = |\vec{a}||\vec{b}|$

$\therefore \vec{a} \cdot \vec{b} = \pm |\vec{a}||\vec{b}| \quad (\because |x| = \pm x)$

$$\cos\theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} = \pm \frac{|\vec{a}||\vec{b}|}{|\vec{a}||\vec{b}|} = \pm 1$$

For $+1 \Rightarrow \theta = 0 \Rightarrow \vec{a} \parallel \vec{b}$

For $-1 \Rightarrow \theta = 180^\circ \Rightarrow \vec{a} \parallel \vec{b}$



14 Suppose $\mathbf{a} \bullet \mathbf{b} = -12$ and $\mathbf{b}$ is a unit vector.

12L

a Explain why $\mathbf{a}$ and $\mathbf{b}$ are not perpendicular.

b Given that $\mathbf{a}$ and $\mathbf{b}$ are parallel, find $|\mathbf{a}|$.

(a) $\vec{a} \cdot \vec{b} = |\vec{a}|\,|\vec{b}| \cos\theta$

$-12 = |\vec{a}|\,|\vec{b}| \cos\theta = |\vec{a}| \cos\theta$

$\neq 0 \quad \neq 0$

$|\vec{a}| \neq 0$ & $\cos\theta \neq 0$

$\therefore$ $\vec{a}$ is not $\perp$ to $\vec{b}$

(b) $\vec{a} \cdot \vec{b} = -12$

$|\vec{a}|\,|\vec{b}| \cos\theta = -12$

$|\vec{b}| = 1$

$|\vec{a}| = \pm 12$

$|\vec{a}| \cos\theta = -12$

$|\vec{a}| \cos\theta = -12$

$\cos\theta = \pm 1$

$|\vec{a}|\,(1) = -12 \Rightarrow |\vec{a}| = -12$

$|\vec{a}|\,(-1) = -12 \Rightarrow |\vec{a}| = 12$

But $|\vec{a}| > 0$

$\therefore$ $\boxed{|\vec{a}| = 12}$

15 Suppose $|\mathbf{c}| = |\mathbf{d}| = \sqrt{5}$. What can be deduced about $\mathbf{c}$ and $\mathbf{d}$ if:

12L

a $\mathbf{c} \bullet \mathbf{d} = 5$ **b** $\mathbf{c} \bullet \mathbf{d} = -5$?

(a) $\vec{c} \cdot \vec{d} = |\vec{c}||\vec{d}| \cos\theta = 5$

$\therefore (\sqrt{5})(\sqrt{5}) \cos\theta = 5$

$5\cos\theta = 5$

$\cos\theta = 1$

$\theta = 0^\circ$

$\therefore \vec{c} \parallel \vec{d}$ & $|\vec{c}| = |\vec{d}|$

$\therefore \vec{c} = \vec{d}$

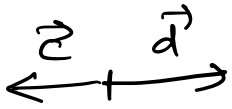
(b) $\vec{c} \cdot \vec{d} = -5$

$|\vec{c}||\vec{d}| \cos\theta = -5$

$(\sqrt{5})(\sqrt{5}) \cos\theta = -5$

$5\cos\theta = -5$

$\cos\theta = -1$



$\therefore \theta = 180^\circ$

$\therefore \boxed{\vec{c} = -\vec{d}}$

16 Find three vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ in space such that $\mathbf{a} \neq 0$ and $\mathbf{a} \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{c}$ but $\mathbf{b} \neq \mathbf{c}$.

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$\vec{a} = i$, $\vec{b} = j$, $\vec{c} = k$

$\vec{a} \neq 0$

$\therefore \vec{a} \cdot \vec{b} = i \cdot j = 0$

$\vec{a} \cdot \vec{c} = i \cdot k = 0$

$\therefore \vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c}$, $\vec{b} \neq \vec{c}$

17 Find any two vectors which are not parallel, but which are *both* perpendicular to $\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$.

12 L

$\begin{pmatrix} 2 \\ -3 \end{pmatrix} \rightarrow \begin{pmatrix} 3 \\ 2 \end{pmatrix}$

Let $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ is $\perp^r$ to $\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$

$\therefore \begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = 0$

let $a=1, b=2, c=c$

$\therefore \begin{pmatrix} 1 \\ 2 \\ c \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = 0 \Rightarrow (1)(1)+(2)(2)+(c)(-1)=0$

$1+4-c=0$

$c=5$

$\therefore \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix}$ is $\perp^r$ to $\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$

Now let $\begin{pmatrix} 2 \\ 2 \\ c \end{pmatrix}$ is $\perp^r$ to $\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$

$\therefore \begin{pmatrix} 2 \\ 2 \\ c \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = 0$

$(2)+(4)-c=0 \Rightarrow c=6$

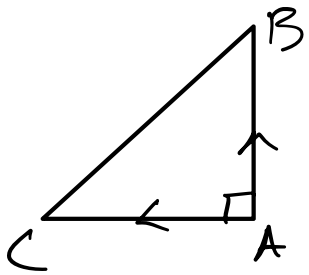
$\therefore \begin{pmatrix} 2 \\ 2 \\ 6 \end{pmatrix}$ is $\perp^r$ to $\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$

Check: $\begin{pmatrix} 2 \\ 2 \\ 6 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = 2+4-6 = 0$ ✓

also $\begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = 1+4-5 = 5-5 = 0$ ✓

18 Consider triangle ABC in which A is $(5, 1, 2)$, B is $(6, -1, 0)$, and C is $(3, 2, 0)$. Using scalar product only, show that the triangle is right angled.

12 L



$$\vec{AB} = \begin{pmatrix} 6-5 \\ -1-1 \\ 0-2 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ -2 \end{pmatrix}$$

$$\vec{BC} = \begin{pmatrix} 3-6 \\ 2-(-1) \\ 0-0 \end{pmatrix} = \begin{pmatrix} -3 \\ 3 \\ 0 \end{pmatrix}$$

$$\vec{AC} = \begin{pmatrix} 3-5 \\ 2-1 \\ 0-2 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ -2 \end{pmatrix}$$

$$\vec{AB} \cdot \vec{BC} = \begin{pmatrix} 1 \\ -2 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ 3 \\ 0 \end{pmatrix} = -3 - 6 + 0 = -8 \neq 0$$

$$\vec{AB} \cdot \vec{AC} = \begin{pmatrix} 1 \\ -2 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 1 \\ -2 \end{pmatrix} = -2 - 2 + 4 = -4 + 4 = 0$$

$\vec{AB} \perp \vec{AC}$

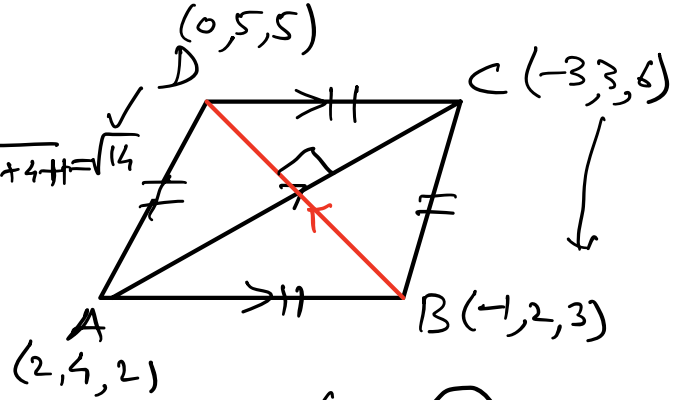
$\therefore$ $\triangle ABC$ is rt. angled at A.

19 A(2, 4, 2), B(−1, 2, 3), C(−3, 3, 6), and D(0, 5, 5) are vertices of a quadrilateral.

12L **a** Prove that ABCD is a parallelogram.

b Find $|\overrightarrow{AB}|$ and $|\overrightarrow{BC}|$. What can be said about ABCD?

c Find $\overrightarrow{AC} \bullet \overrightarrow{BD}$ and discuss your answer.



(a) $\overrightarrow{AB} = \begin{pmatrix} -1-2 \\ 2-4 \\ 3-2 \end{pmatrix} = \begin{pmatrix} -3 \\ -2 \\ 1 \end{pmatrix} \Rightarrow |\overrightarrow{AB}| = \sqrt{9+4+1} = \sqrt{14}$ ✓

$\overrightarrow{DC} = \begin{pmatrix} -3-0 \\ 3-5 \\ 6-5 \end{pmatrix} = \begin{pmatrix} -3 \\ -2 \\ 1 \end{pmatrix}$

$\overrightarrow{AB} \parallel \overrightarrow{DC}$

$\overrightarrow{AD} = \begin{pmatrix} 0-2 \\ 5-4 \\ 5-2 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix}$, $\overrightarrow{BC} = \begin{pmatrix} -3+1 \\ 3-2 \\ 6-3 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} \Rightarrow \sqrt{4+1+9} = \sqrt{14}$ ✓

$\overrightarrow{AD} \parallel \overrightarrow{BC}$

$\Rightarrow$ ABCD is a parallelogram (Rhombus)

(b) $|\overrightarrow{AB}| = |\overrightarrow{BC}| = \sqrt{14}$

$\therefore$ ABCD is a Rhombus.

(c) $\vec{AC} = \begin{pmatrix} -3-2 \\ 3-4 \\ 6-2 \end{pmatrix} = \begin{pmatrix} -5 \\ -1 \\ 4 \end{pmatrix}$

$\vec{BD} = \begin{pmatrix} 0+1 \\ 5-2 \\ 5-3 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$

$\therefore \vec{AC} \cdot \vec{BD} = (-5)(1) + (-1)(3) + (4)(2) = -5-3+8 = 0$

$\vec{AC} \perp \vec{BD}$

$\vec{AB} \cdot \vec{BC} = \begin{pmatrix} -3 \\ -2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} = 6-2+3 = 7 \neq 0$

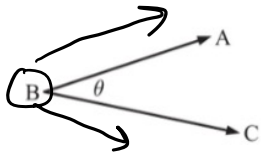
$\therefore$ ABCD is a <u>rhombus</u>.

20 Find the angle ABC of triangle ABC for A(3, 0, 1), B(−3, 1, 2), and C(−2, 1, −1).

12L

Hint: To find the angle at B, use $\overrightarrow{BA}$ and $\overrightarrow{BC}$.

What angle is found if $\overrightarrow{BA}$ and $\overrightarrow{CB}$ are used?

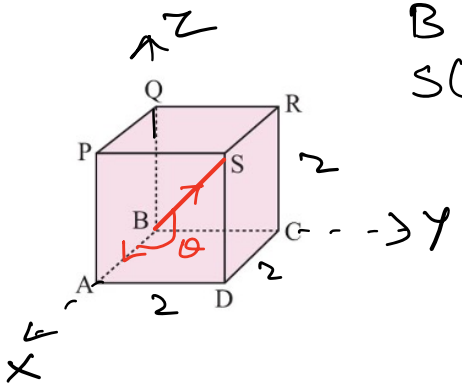


H.W.

$$\cos\theta = \frac{|\vec{a}\cdot\vec{b}|}{|\vec{a}|\,|\vec{b}|}$$

21 The cube alongside has edges of length 2 cm.
12L Find, using vector methods, the measure of:

a $A\hat{B}S$ **b** $R\hat{B}P$ **c** $P\hat{B}S$



B (0, 0, 0)
S (2, 2, 2) ✓
A (2, 0, 0)
C (0, 2, 0)
Q (0, 0, 2)
zx-plane → P (2, 0, 2)
yz-plane → R (0, 2, 2)
xy-plane → D (2, 2, 0)

(a) $\vec{BA} = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}$

$\vec{BS} = \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix}$

$\therefore \quad \vec{BA} \cdot \vec{BS} = (2)(2) + 0 + 0 = 4$

$|\vec{BA}| = \sqrt{2^2} = 2$

$|\vec{BS}| = \sqrt{4+4+4} = \sqrt{12} = 2\sqrt{3}$

$\therefore \quad \cos\theta = \dfrac{\vec{BA} \cdot \vec{BS}}{|\vec{BA}|\,|\vec{BS}|} = \dfrac{4}{(2)(2\sqrt{3})} = \dfrac{1}{\sqrt{3}}$

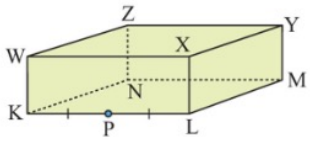
$\cos\theta = \dfrac{1}{\sqrt{3}} \Rightarrow \boxed{\theta = 54.7^\circ}$

HW.

22 [KL], [LM], and [LX] are 8, 5, and 3 units long respectively. P is the midpoint of [KL]. Find, using vector methods, the measure of:

12 L

a $Y\hat{N}X$ **b** $Y\hat{N}P$



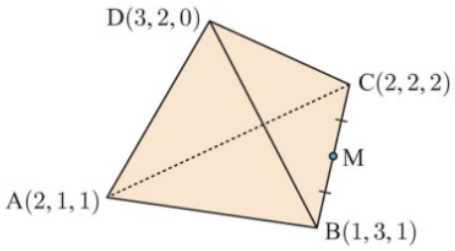
H.W.

23 Consider the tetrahedron ABCD.

12L

a Find the coordinates of M.

b Find the measure of $D\widehat{M}A$.



End of Lesson